CLAIMPROMOTEBENCH: SYNTHETIC POLICY-CONSTRAINED EVALUATION OF EVIDENCE-STATE TRANSITIONS

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ABSTRACT

Producing evidence does not determine which claim-state transition it licenses. We evaluate this intervening decision as synthetic policy conformance: given a claim, typed evidence, current state, target, and fixed author-defined policy, return `ALLOW`, `HOLD`, `DEMOTE`, or `ESCALATE`. CLAIMPROMOTEBENCH uses controlled pairs and counts every service or output failure as incorrect. In a preregistered hidden study of 300 items and 150 pairs, one Qwen3.7-Max service configuration returned 272/300 valid decisions and 123/150 complete pairs, yielding .8600/.7533 end-to-end item/pair accuracy versus .9485/.9187 conditional accuracy at .9067/.8200 coverage. End-to-end action recall was .9533/.8209/.0000/.6250 in the order above. A precommitted public challenge balanced 90 `ALLOW` members with 30 per other action across 15 factors. Qwen and Flash routes reached .7333/.5333 and .5000/.1444 end-to-end item/pair accuracy; conditional values were .8919/.7500 and .6164/.2131. Their full confusion matrices expose `DEMOTE` recall of 4/30 and 0/30; each configuration has a zero-accuracy factor. The old Flash lane documents 300/300 HTTP 401 authorization failures on its old frozen service interface and supplies no model-efficacy estimate; the later route establishes only execution under a different stack. This preregistered diagnostic case study is neither a causal ablation nor evidence of general research-agent ability. The deterministic executor is the definitional policy ceiling, and the Naive Bayes comparator remains a weak baseline.

1 INTRODUCTION

A valid artifact licenses only the scope it checks. A finite computation supports its registered finite domain, while a proof certificate supports its exact encoded statement in a pinned environment. Independent reproduction changes provenance rather than scope. A system that collapses these distinctions can assign a claim status stronger than the evidence warrants.

We isolate the decision between evidence production and claim-state assignment:

> *Given a claim, typed evidence, a current state, and a requested transition, which action conforms to a fixed, machine-checkable policy?*

The benchmark treats this question as synthetic policy conformance under an authored operational definition. Oracle agreement therefore measures behavior relative to that definition, not scientific truth or community consensus. The task excludes open-ended discovery, experimental design, literature synthesis, and general research-agent competence.

CLAIMPROMOTEBENCH records research status, computational support, formal certification, and independent review on separate state axes. A controlled pair fixes the claim and requested transition while changing one registered prerequisite. We score `ALLOW`, `HOLD`, `DEMOTE`, and `ESCALATE`, retaining unusable responses in the end-to-end denominator. Coverage measures whether a service returns a valid decision; conditional accuracy scores valid responses; end-to-end item and pair accuracy score every scheduled case.

Our contributions are:

1. a synthetic benchmark that maps typed evidence and requested state transitions to four actions under a fixed executable policy;
2. failure-aware item and pair estimands, applied in a preregistered hidden diagnostic and a precommitted action-balanced public challenge; and
3. a replayable evidence chain that separates service output from a definitional policy ceiling and produces privacy-filtered reviewer artifacts.

2 Related Work

Behavioral and contrastive evaluation. CheckList organizes behavioral tests around capabilities and expected invariances (Ribeiro et al., 2020), while contrast sets probe local decision boundaries through small, meaningful input changes (Gardner et al., 2020). Metamorphic testing similarly checks relations among transformed inputs when a conventional test oracle is difficult to apply (Chan et al., 1998). ClaimPromoteBench uses authored policy labels and controlled pairs: most pairs intentionally change the action, while separate metamorphic checks preserve the action under identifier renaming, evidence reordering, and irrelevant-evidence insertion. These tests characterize selected local behaviors; they are not an estimate of natural-distribution performance.

Structured output and selective prediction. Incremental parsing and grammar-constrained decoding can prevent structurally invalid generations (Scholak et al., 2021; Geng et al., 2023). They cannot guarantee that a structurally valid action satisfies cross-field semantics or the evidence policy. Selective classification studies risk–coverage trade-offs and reject options (Geifman & El-Yaniv, 2017; 2019); recent abstention benchmarks expose continuing failures on unanswerable tasks (Kirichenko et al., 2025). We therefore report coverage and conditional correctness together, while end-to-end metrics count every absent or invalid decision as incorrect. `HOLD` is a policy action about current evidence, whereas service failure is recorded as no valid decision.

Explicit rule following and formal specifications. RuleArena evaluates long-context natural-language rule application for airline baggage fees, NBA transactions, and tax regulations, including logical and numerical reasoning (Zhou et al., 2025). OSVBench instead asks models to generate formal specifications for operating-system verification (Li et al., 2026). ClaimPromoteBench neither answers domain questions nor generates specifications: it applies a complete machine-readable evidence policy to a typed ledger and scores a four-way state-transition action. Its oracle is an authored synthetic definition, not an external judgment of scientific truth.

Scientific evidence and research-agent evaluation. CLAIM-BENCH, DeepFact, and ClaimDB evaluate scientific claim extraction or factual verification (Javaji et al., 2025; Huang et al., 2026; Theologitis et al., 2026). ScienceAgentBench, CORE-Bench, and POPPER evaluate executable research tasks, computational reproduction, or hypothesis falsification (Chen et al., 2025; Siegel et al., 2025; Huang et al., 2025). SeekBench, Omni-Decision, EviGraph, and ClaimLedger emphasize grounding or explicit evidence maintenance (Shao et al., 2026; Ma et al., 2026; Ren et al., 2026; Elliman, 2026). Unlike these broader tasks, ClaimPromoteBench fixes an authored policy and scores only the status-transition decision after evidence registration.

3 Task and Evidence Flow

Each item is $x = (c, E, s, \tau, t)$: a claim c, typed evidence E, current state s, requested target τ, and frozen evaluation time t. The state has four axes:

$$\begin{aligned} \texttt{research_status} &\in \{\texttt{PROPOSED}, \texttt{OPEN}, \texttt{RESOLVED}\}, \\ \texttt{computation_status} &\in \{\texttt{NONE}, \texttt{COMPUTATIONAL}\}, \\ \texttt{formal_status} &\in \{\texttt{NONE}, \texttt{FORMALLY_CERTIFIED}\}, \\ \texttt{review_status} &\in \{\texttt{UNREVIEWED}, \texttt{INDEPENDENTLY_VERIFIED}\}. \end{aligned}$$

The public policy checks dependencies, scope, freshness, provenance, runtime completion, contradictions, statement identity, and artifact integrity. Its deterministic oracle $\pi_P(x)$ evaluates the following rows from top to bottom; the first matching row fixes both the action and proposed state.

Action	First-match condition	Proposed state
`DEMOTE`	A currently elevated axis has an absent, broken, stale, mismatched, or policy-nonconforming recorded basis.	Lower each affected axis.
`ESCALATE`	A fresh, identity-matched, independently produced contradiction is open and passes its runtime and verifier checks.	Keep current state.
`HOLD`	The requested promotion lacks a complete valid dependency closure or any required scope, freshness, runtime, verifier, provenance, or axis-specific check.	Keep current state.
`ALLOW`	None of the preceding conditions holds.	Accept target state.

The model receives the complete versioned policy contract, a canonical JSON evidence ledger, and passive verifier checks that report raw facts but no transition decision. Action precedence is `DEMOTE` > `ESCALATE` > `HOLD` > `ALLOW`; reason codes follow the policy's published total order. The response must be one JSON object with exactly six fields: `action`, `proposed_state`, `p_admissible`, `used_evidence_ids`, `reason_codes`, and `missing_evidence_ids`. No rationale text is requested.

Let $v(x)$ indicate a syntactically, schema-, and semantically valid response and let $\hat{y}(x)$ be its action. End-to-end item correctness is

$$C_{\text{item}}(x) = \mathbb{1}[v(x) = 1]\mathbb{1}[\hat{y}(x) = \pi_P(x)].$$

For controlled pair $i = (x_i^a, x_i^b)$, $C_{\text{pair}}(i) = C_{\text{item}}(x_i^a)C_{\text{item}}(x_i^b)$. Decision coverage is the fraction with $v = 1$; pair coverage is the fraction with both members valid. Conditional accuracies divide correct items or pairs by those coverage denominators. End-to-end accuracies divide by all scheduled items or pairs.

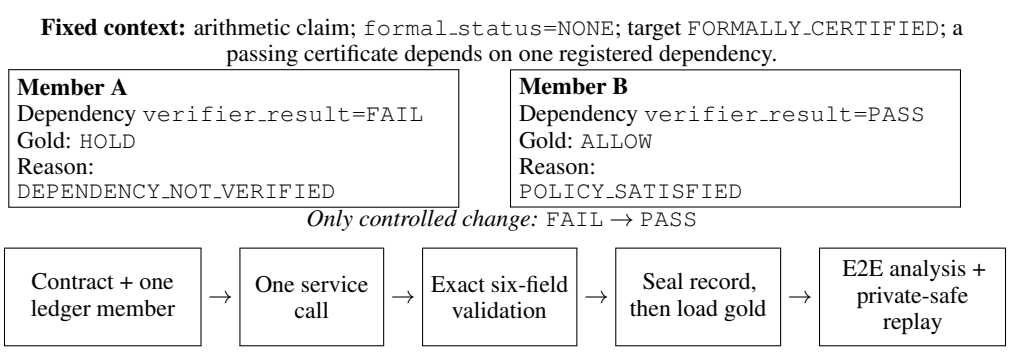


Figure 1: A concrete controlled pair and the evidence flow. The public challenge pair changes only `/evidence/1/verifier_result`; fixed identifiers and hashes are omitted from the diagram. End-to-end analysis retains every terminal failure.

Within a controlled pair, only registered fields relevant to one factor change. The illustrated member changes a dependency check from `FAIL` to `PASS`; other mappings send a missing dependency to `HOLD`, a current-state binding failure to `DEMOTE`, and an active contradiction to `ESCALATE`. Separate invariance tests rename identifiers, reorder evidence, or add irrelevant records and require both the oracle and rendered prompt semantics to remain unchanged.

Table 1: Factor inventory, grouped by policy family. Labels humanize the machine-readable factor identifiers; counts are 20 for r6 and 15 for the challenge.

Family	r6 factors	Public-challenge factors
Dependency	dependency statement, dependency verifier, direct dependency, transitive dependency	dependency verifier
Formal identity	claim identity, statement identity, trust base	—
Numerical, current basis	domain, precision, preregistration, repetitions	current computation binding, staleness, verifier; current review environment, identity; repetitions
Provenance	cold start, review environment, reviewer identity	reviewer identity
Runtime	run completion, tool exit, verifier execution	tool exit
Freshness, contradiction	age, contradiction, current binding	age; contradiction claim identity, freshness, reviewer identity, status, verifier execution

4 Benchmark and Evaluation Design

4.1 Synthetic benchmark

Version 1 contains 200 pairs (400 items): 50 public-development pairs and 150 hidden pairs over the 20 factors in Table 1. Numerical cases recompute registered finite-grid values and scope, while formal cases compile generated Lean sources under a pinned toolchain. These checks establish internal policy consistency, not external scientific validity.

The fresh r6 realization contains 150 hidden pairs and 300 unique prompts from the same ontology and mostly the same template family. A preregistered construction commitment and request schedule preceded inference; all 600 terminal cells for two service lanes were sealed before gold loading. Each item received one call with zero client retries. The gold distribution is imbalanced: 150 `ALLOW`, 134 `HOLD`, eight `DEMOTE`, and eight `ESCALATE` items. We also froze a four-class multinomial Naive Bayes classifier ($\alpha = 1$) trained on identifier-normalized unigram and bigram prompt counts from the 100 public-development items.

4.2 Balanced public robustness challenge

The separately precommitted public challenge tests action asymmetry over the 15 factors in Table 1, with six pairs per factor. Its 90 pairs contain 90 `ALLOW` members and 30 members for each of `HOLD`, `DEMOTE`, and `ESCALATE`. We froze the dataset, labels, prompts, adapters, analysis, and execution contract before inference. Each service route made 180 calls with zero client retries. This public, preregistered robustness challenge tests a known policy and public synthetic construction; it does not estimate hidden-test or natural-distribution generalization.

The two routes differ in model and service stack, including output-constraining behavior. We report descriptive service-configuration outcomes because the design cannot identify the effect of model weights, provider, schema mode, or any single component.

5 Results

5.1 Coverage, conditional accuracy, and end-to-end accuracy

The r6 Qwen configuration returned 272/300 valid decisions and 123/150 complete pairs (Table 2). Conditional item/pair accuracy was .9485/.9187; retaining all scheduled failures reduced end-to-end accuracy to .8600/.7533. Fourteen HTTP 500 responses, one HTTP 503 response, ten timeouts, and three semantic-invalid outputs remain incorrect. The frozen Naive Bayes comparator scored .6133 item and .2267 pair accuracy. It checks a low-cost learned path but is a weak baseline, not a representative competitive comparator.

Table 2: Coverage and accuracy. Conditional values use only valid decisions or complete pairs; end-to-end (E2E) values retain all scheduled failures. "Old" identifies the frozen r6 interface. N/E denotes undefined conditional correctness and non-evaluable model efficacy because no model output was returned.

Configuration	Valid	E2E item	Cond. item	Pair cov.	E2E pair	Cond. pair
r6 Qwen3.7-Max	272/300	.8600	.9485	.8200	.7533	.9187
r6 public-dev text NB	300/300	.6133	.6133	1.0000	.2267	.2267
r6 old Flash interface	0/300	.0000	N/E	.0000	.0000	N/E
Public challenge Qwen	148/180	.7333	.8919	.7111	.5333	.7500
Public challenge Flash	146/180	.5000	.6164	.6778	.1444	.2131

The r6 Flash lane returned 300/300 HTTP 401 authorization failures on the old frozen service interface and no model output. Its scheduled end-to-end scores are 0/300 items and 0/150 pairs; conditional correctness, model efficacy, and the registered second confirmation are non-evaluable. The later challenge route returned model outputs under a different service stack, establishing only that its own path executed. It neither repairs r6 nor supports a strict causal comparison.

On the balanced challenge, Qwen returned 148/180 valid decisions and 64/90 complete pairs; Flash returned 146/180 valid decisions and 61/90 complete pairs. Their end-to-end item/pair accuracies were .7333/.5333 and .5000/.1444, while conditional accuracies were .8919/.7500 and .6164/.2131. Qwen recorded 31 transport failures and one semantic failure; Flash recorded 34 semantic failures. The larger pair-level gaps show why conditional correctness cannot replace coverage or the end-to-end estimand.

5.2 Factor-level pair outcomes

Factor-level pair results vary sharply (Table 3). For r6 Qwen, dependency statement scores 0/7 and current binding scores 0/8, with pair coverage of 2/7 and 6/8; seven factors are perfect. Challenge Qwen has three zero-accuracy factors and two perfect factors. Challenge Flash has 11 zero-accuracy factors and none that are perfect. Supports of six to eight pairs make these descriptive breakdowns, not separate confirmatory tests.

Table 3: Factor-level pair coverage and correctness. Each entry is complete/correct pairs; E2E accuracy is correct/all pairs. Parentheses give r6 support; every challenge factor has six pairs.

r6 Qwen: 20 factors

Factor (pairs)	Complete/correct	Factor (pairs)	Complete/correct
Age (8)	7/7	Claim identity (8)	7/7
Cold start (8)	7/7	Contradiction (8)	5/5
Current binding (8)	6/0	Dependency statement (7)	2/0
Dependency verifier (7)	7/7	Direct dependency (7)	6/5
Domain (7)	5/5	Precision (7)	7/7
Preregistration (7)	7/7	Repetitions (7)	7/7
Review environment (8)	7/7	Reviewer identity (8)	5/5
Run completion (8)	7/7	Statement identity (8)	4/3
Tool exit (7)	7/7	Transitive dependency (7)	7/7
Trust base (8)	6/6	Verifier execution (7)	7/7

Public challenge: 15 factors

Factor	Qwen complete/correct	Flash complete/correct
Age	5/5	6/0
Contradiction claim identity	5/5	0/0
Contradiction freshness	3/3	4/0
Contradiction reviewer identity	3/3	2/0
Contradiction status	6/6	4/4
Contradiction verifier execution	3/3	1/0
Current computation binding	4/0	6/0
Current computation stale	5/0	6/0
Current computation verifier	4/2	4/0
Current review environment	3/0	6/0
Current review identity	3/1	6/0
Dependency verifier	5/5	5/5
Repetitions	5/5	4/1
Reviewer identity	4/4	4/0
Tool exit	6/6	3/3

5.3 Action recall and full confusion matrices

Table 4 shows end-to-end recall, so no-valid-decision cells count as misses. R6's eight-item DEMOTE and ESCALATE supports are too small for a uniform four-action claim. The balanced challenge supplies 30 members per non-ALLOW action and preserves the same qualitative DEMOTE weakness: Qwen recovers 4/30 and Flash 0/30.

Table 4: End-to-end recall by gold action; no-valid cells are misses. Gold supports are 150/134/8/8 for r6 and 90/30/30/30 for each challenge lane. Macro-F1 averages all four actions with zero division set to zero.

Configuration	ALLOW	HOLD	DEMOTE	ESCALATE	Bal. recall	Macro-F1
r6 Qwen3.7-Max	.9533	.8209	.0000	.6250	.5998	.6265
Public challenge Qwen	.8778	.8667	.1333	.7667	.6611	.7004
Public challenge Flash	.7556	.3000	.0000	.4333	.3722	.3928

Table 5 gives every end-to-end confusion cell. The final column distinguishes absence of a valid decision from any policy action. R6 Qwen maps all eight gold DEMOTE items to ALLOW. On the public challenge, Qwen's 20 valid gold-DEMOTE responses split across all four actions, while Flash never predicts DEMOTE. Flash also maps 13 gold-HOLD items to ALLOW and loses 14/30 gold-ESCALATE items to invalid responses.

Table 5: Full confusion matrices. Rows are gold actions; action columns are valid predictions; "No valid" includes service, syntax, schema, and semantic failures.

Configuration	Gold	ALLOW	HOLD	DEMOTE	ESCALATE	No valid
r6 Qwen	ALLOW	143	0	0	0	7
	HOLD	4	110	0	2	18
	DEMOTE	8	0	0	0	0
	ESCALATE	0	0	0	5	3
Challenge Qwen	ALLOW	79	0	0	0	11
	HOLD	0	26	0	0	4
	DEMOTE	8	4	4	4	10
	ESCALATE	0	0	0	23	7
Challenge Flash	ALLOW	68	4	0	8	10
	HOLD	13	9	0	0	8
	DEMOTE	25	3	0	0	2
	ESCALATE	0	3	0	13	14

5.4 Definitional policy ceiling

The deterministic policy executor reads the same model-visible item and public policy that define the labels. It reproduces 300/300 r6 items and 150/150 pairs, and all 360 scheduled public-challenge item cells and 180 pair cells. This perfect agreement is true by definition of the evaluation target. We call it a *definitional policy ceiling*; it is neither a learned baseline nor independent evidence that the authored policy is scientifically correct. Post-generation gates driven by the same executor are implementation checks and provide no evidence of learned model competence.

6 Discussion

What r6 supports. The preregistered r6 diagnostic supports one time-stamped Qwen service configuration on a fresh hidden realization of the same synthetic ontology. It handled 258/300 items and 113/150 pairs correctly end to end, with .9067 decision coverage and .8200 pair coverage. Its .9485/.9187 conditional item/pair accuracy answers a narrower question: correctness after a usable response exists. Neither estimand measures broad research ability or naturalistic scientific judgment.

Action balance changes the picture. R6 is dominated by ALLOW and HOLD, and its 0/8 DEMOTE recall precludes a uniform four-action claim. The balanced challenge gives 30 cases to each non-ALLOW action; Qwen recovers 4/30 DEMOTE cases and Flash recovers none. Every configuration

also has a zero-accuracy factor. The confusion matrices distinguish the remaining errors: Qwen mixes valid demotions across all four actions, while Flash often substitutes `ALLOW` for `HOLD` or `DEMOTE`. Action recall, factor breakdowns, and pair accuracy expose failures that aggregate item accuracy hides.

Service reliability is part of end-to-end behavior. Every transport, syntax, schema, and semantic failure is a failed scheduled decision. Conditional reporting separates response validity from policy correctness, but it cannot replace the end-to-end estimand. The old r6 Flash lane documents 300/300 authorization failures on its frozen service interface and no model output. The later Flash route shows only that a different path executed. Because both the service stack and model configuration changed, the Qwen–Flash gap is descriptive and is not a causal ablation.

Comparators and enforcement. The public-development Naive Bayes model gives a reproducible low-cost reference, but its .6133 item accuracy, .2267 pair accuracy, .3284 balanced recall, and .3124 macro-F1 make it a weak baseline. Stronger learned comparators remain needed. The deterministic executor reaches perfect agreement because it defines the gold actions. It is a definitional policy ceiling, and treating it as learned model evidence would be circular.

7 LIMITATIONS

The benchmark is synthetic and policy-constrained. R6 reduces post hoc adaptation through a fresh hidden realization, but it retains the same ontology and mostly the same template family. The public, precommitted challenge tests an action-balanced construction rather than hidden-test generalization. Its 90 pairs and 15 factors broaden action coverage, although six pairs per factor cannot support precise factor-specific estimates. The full factor matrix is diagnostic: zero cells identify failures but do not establish stable factor rankings.

Coverage remains a material limitation. R6 loses 28/300 decisions and 27/150 complete pairs; the public challenge loses 32/180 Qwen decisions and 34/180 Flash decisions. Conditional accuracies of .9485/.9187, .8919/.7500, and .6164/.2131 must therefore be read beside their end-to-end counterparts. Thirty challenge cases per non-`ALLOW` action still support only coarse recall estimates, including the observed 4/30 and 0/30 `DEMOTE` results.

The evidence covers a few time-stamped service configurations and one call per item. Zero retries define the estimand but do not characterize retry policies, sampling variance, fixed model weights, or alternate providers. The old and new Flash routes differ enough that later executability cannot identify why the former returned 401 or what its r6 model accuracy would have been. No strict causal provider, model, or structured-output comparison follows.

The policy and labels are authored operational definitions. There is no human annotation study, external policy consensus, agreement measure, or validation against downstream scientific outcomes. Before any CLAIMPROMOTEBENCH call, we froze and reviewed 12 pointer-only CORE-Bench candidates; none contained the time-stamped artifacts, dependency chain, runtime status, and provenance needed for a transition. We therefore created zero naturalistic benchmark cases and made zero model calls. No human annotators or agreement statistic are claimed. Naturalistic transfer remains open, as do stronger learned baselines, independent policy assessment, and downstream utility.

8 REPRODUCIBILITY AND ARTIFACT POLICY

The repository binds policies, generators, datasets, prompts, terminal records, gold-loading order, analyses, and receipts. R6 freezes its construction and 600-request schedule before inference and seals every terminal record before gold loading. The public challenge freezes its dataset, action balance, prompt packet, adapters, and analysis before inference. Paper construction does not rerun either completed singleton study.

A post-r6 audit found provider hidden-thinking blocks in 275 source envelopes; scoring had used only final response text and parsed decisions. The reviewer derivative excludes those envelopes through a text-only allowlist and verifies source-to-derived hashes and scientific-payload equivalence. The anonymous bundle contains replayable item records, policy and prompt bindings, executable checks,

formal checks, and privacy receipts. Its manifest derives the current file count from the generated tree. Fail-closed scans reject author identity, repository or workflow identifiers, provider response identifiers, hidden reasoning, and unknown record fields.

9 AI Assistance Disclosure

Generative AI tools assisted literature search, planning, code and benchmark drafting, debugging, analysis checks, and manuscript editing. The author chose the research question, defined the policy and protocol, made the study and analysis decisions, verified the benchmark, results, and references, and is responsible for the manuscript. Quantitative results come from validated study artifacts. AI systems are neither authors nor human annotators.

10 Conclusion

ClaimPromoteBench isolates policy-constrained evidence-state transitions from broader research-agent tasks. The hidden r6 Qwen configuration reached .8600/.7533 end-to-end item/pair accuracy at .9067/.8200 coverage; conditional accuracy was .9485/.9187. On the action-balanced public challenge, Qwen and Flash reached .7333/.5333 and .5000/.1444 end to end, with conditional values of .8919/.7500 and .6164/.2131. Their full confusion matrices show `DEMOTE` recall of only 4/30 and 0/30, and every configuration has a zero-accuracy factor. These results support a configuration-specific diagnostic under an authored synthetic policy, not uniform policy competence or general research ability.

All unusable responses remain in the denominator. The old Flash lane documents 300/300 authorization failures on its frozen service interface and no model output; the later route establishes only executability under a different stack. Their outcomes do not form a causal ablation. The deterministic executor is a definitional policy ceiling, and the Naive Bayes comparator remains weak. Source-grounded naturalistic cases, stronger learned baselines, repeated configurations, and independent policy assessment are the next tests.

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